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What is a unit test?

A unit test is a small program that verifies that a minimal unit of code (typically a method or a class) functions correctly in isolation.

What is JUnit?

JUnit is an open source framework for writing and running automated unit tests in Java. It provides:

- Annotations to mark and organize tests (@Test, @BeforeEach, etc).
- Assertions to verify results (assertEquals, assertThrows, etc.)
- A test runner that discovers tests, executes them, and reports which ones passed or failed

How is it implemented Java?

1. Add dependency
2. Folder structure
3. Class to test
4. Test class
5. Run

1. With Maven (pom.xml):

```
<dependencies>
  <dependency>
    <groupId>org.junit.jupiter</groupId>
    <artifactId>junit-jupiter</artifactId>
    <version>5.10.2</version>
    <scope>test</scope>
  </dependency>
</dependencies>
```

Notes

```
<build>
  <plugins>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-surefire-plugin</artifactId>
      <version>3.2.5</version>
    </plugin>
  </plugins>
</build>
```

With Gradle (build.gradle):

```
dependencies {
    testImplementation 'org.junit.jupiter:junit-jupiter:5.10.2'
}
test {
    useJUnitPlatform()
}
```

Step ② : Project structure

```
my-project
├── pom.xml
└── src/
    ├── main/java/com/example/
    │   └── Calculator.java
    └── test/java/com/example/
        └── CalculatorTest.java
```

Escaneado con CamScanner